

Report

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1 Introduction

This project is aimed to develop a high-performance automated classifier capable of distinguishing between Real and Fake news snippets. By leveraging Sentence Embeddings (all-MiniLM-L6-v2) and machine learning algorithms (specifically Logistic Regression and Random Forest), the project aims to move beyond simple keyword matching. The goal is to capture deep semantic patterns and stylistic nuances in text to provide a robust defense against digital misinformation.

2 Methodology

2.1 Dataset Creation

The dataset for this project was constructed by integrating two distinct sources from Kaggle: the Fake and Real News Dataset by Clément Bisillon[1] and the News Detection dataset by Nitish Jolly[2]. To ensure data quality and model robustness, the preprocessing pipeline utilized Python and pandas to integrate multiple CSV files into a unified structure. Furthermore, labels were standardized into a consistent binary format, defining 0 as "Real" and 1 as "Fake." The finalized dataset is hosted on the Hugging Face[3].

2.2 Embeddings & Model Training

This project utilizes sentence embeddings to capture deep contextual meaning. Specifically, the all-MiniLM-L6-v2 model from the sentence-transformers library was employed to transform news text into normalized 384-dimensional dense vectors, thereby preserving semantic relationships within a high-dimensional space. To evaluate different algorithmic approaches, two classifiers were trained: Logistic Regression, serving as a high-efficiency linear baseline, and Random Forest, utilized to capture potential non-linear patterns within the embeddings. Ultimately, the optimized model was deployed as an interactive web application via the Gradio framework and is currently hosted on Hugging Face Spaces[3], enabling users to receive real-time predictions and confidence scores. All source code, including the preprocessing scripts and model training pipelines, is publicly available in the project repository on GitHub[4].

3 Evaluation & Results

The experimental results demonstrate exceptional performance, with the optimized model achieving an accuracy of 96.21% and an F1-score of 0.9648. In a comparative analysis of classifiers, Logistic Regression showed remarkably high efficiency and precision, slightly outperforming Random Forest in handling the high-dimensional embedding space. The primary reason for such robust performance lies in the use of Sentence Embeddings, which provide a significant advantage over traditional bag-of-words models. By mapping text into a dense semantic space, the embeddings successfully capture the nuanced linguistic patterns and subtle stylistic differences between credible reporting and deceptive misinformation, allowing the classifiers to distinguish the two classes with high confidence.

4 Reflection on AI Collaboration

This project was developed through a synergistic collaboration between me and AI coding assistants (e.g. Cursor). AI tools played a pivotal role in accelerating the development cycle, particularly in debugging environment-specific issues. A notable example was resolving a binary compatibility conflict caused by the release of NumPy 2.0. By interacting with the AI, I was able to quickly identify that the pre-compiled tokenizers wheels were incompatible with the new NumPy version, leading to a swift resolution by implementing specific version constraints in "requirements.txt".

A key insight from this collaboration is that the quality of the AI's output is strictly bounded by the quality of the user's prompts. To achieve a 96.2% accuracy, I had to provide structured, context-rich instructions that defined clear constraints and desired outcomes. Vague prompts often led to hallucinations, whereas well-engineered prompts acted as a robust specification for the AI to follow.

References

- [1] C. Bisailon. (2020). Fake and Real News Dataset. Kaggle.
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- [2] N. Jolly. (2020). News Detection (Fake or Real) Dataset. Kaggle.
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- [3] Y. Hanqi. (2026). Fake-Real-News Dataset. Hugging Face. <https://huggingface.co/datasets/yahuqiao/fake-real-news>
- [4] Y. Hanqi. (2026). *Fake News Detection Project*. GitHub Repository. https://github.com/yhanqi0611-create/fake_news_detection